



NBA SHOT TRENDS

Did Stephen Curry Really Change
the Game?

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01 Goal & Objectives



Objectives

- ❑ **Explore NBA shooting trends over time:** Analyze league-wide shot selection data over two decades.
- ❑ **Isolate Stephen Curry's most impactful season (2015–16):** Investigate Curry's record-breaking season to assess significance in shifting league norms.
- ❑ **Examine Potential influence on league-wide behavior:** Determine whether Curry's increased three-point volume and efficiency correlate with broader changes in how teams and players approach shot selection.
- ❑ **Visualize changes in shot distribution across seasons:** Create interactive visualizations to communicate.
- ❑ **Intended audience:** Basketball fans

02

Dataset



- ❑ **NBA Shots dataset (Kaggle):** shot-by-shot data from NBA games from the 2004–05 season through the 2023-24 season.
- ❑ Thousands of individual shot attempts with variables: such as **season**, **player_name**, **team_name**, **shot_made**, **shot_distance**, **shot_type**, etc...
- ❑ Features allow us to conduct a spatial analysis of shooting behavior as well with variables: **loc_x** and **loc_y**
 - ❑ Allows mapping on basketball court display
- ❑ **Overall:** Allows isolation of players, periods, and shots and analysis of how spatial behavior changes



03

Implementation & Design



- Split Shiny dashboard into three tabs:

- Shot Heatmap
- Comparing Players
- NBA Metrics (Team Level)

- Simplicity when examining graphs

- Ease of navigation

- Use NBA color scheme (Red, White, Blue)



NBA Dashboard

Shot Heatmap

Player Comparison

NBA Metrics

Player 1:

Kobe Bryant

Year 1:

2013

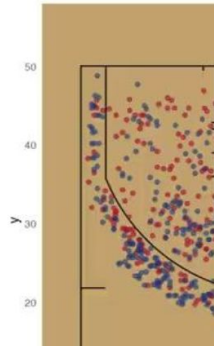
Player 2:

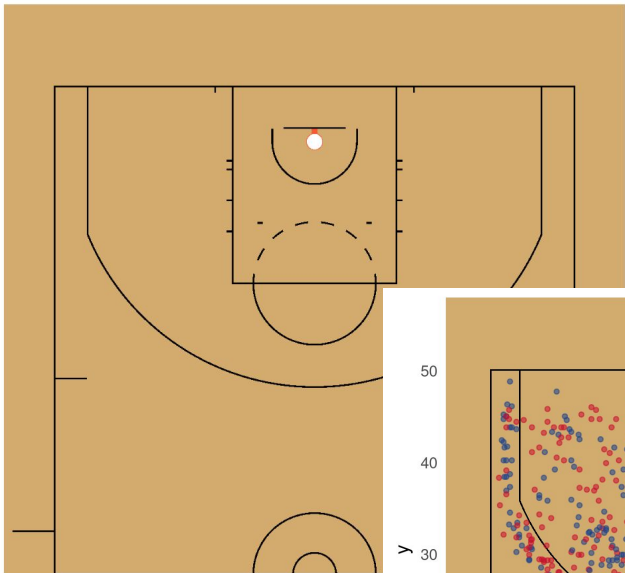
Stephen Curry

Year 2:

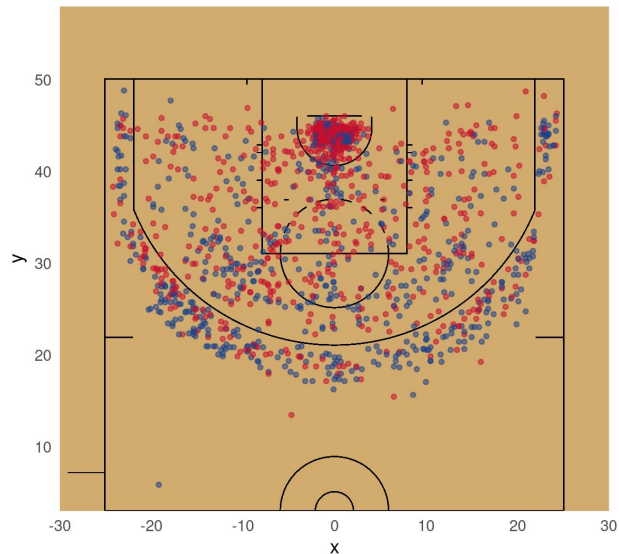
2013

Compare Player Shot Locations





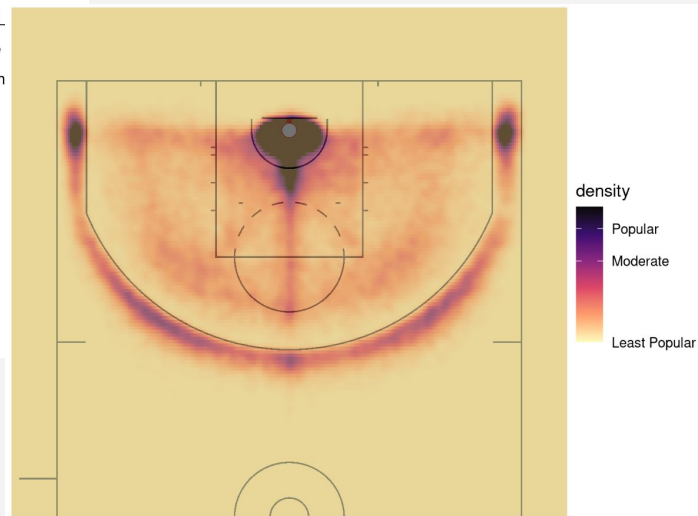
Comparing Players



Shot Heatmap

PLAYER_

- Kobe
- Steph



04

Demo



NBA Dashboard

Shot Heatmap

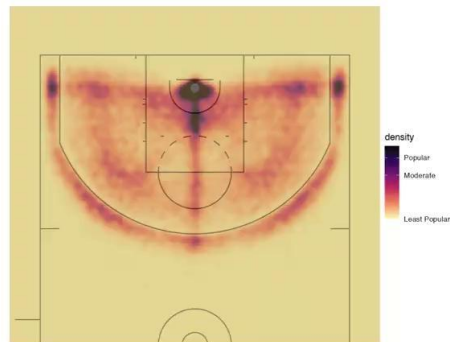
Player Comparison

NBA Metrics

Select Year:

2004

Shot Density Heatmap



NBA Dashboard

Shot Heatmap

Player Comparison

NBA Metrics

Player 1:

Kobe Bryant

Year 1:

2013

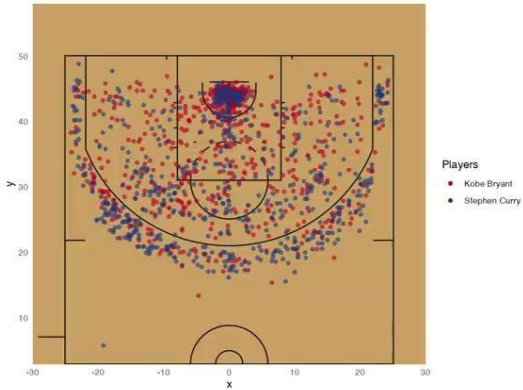
Player 2:

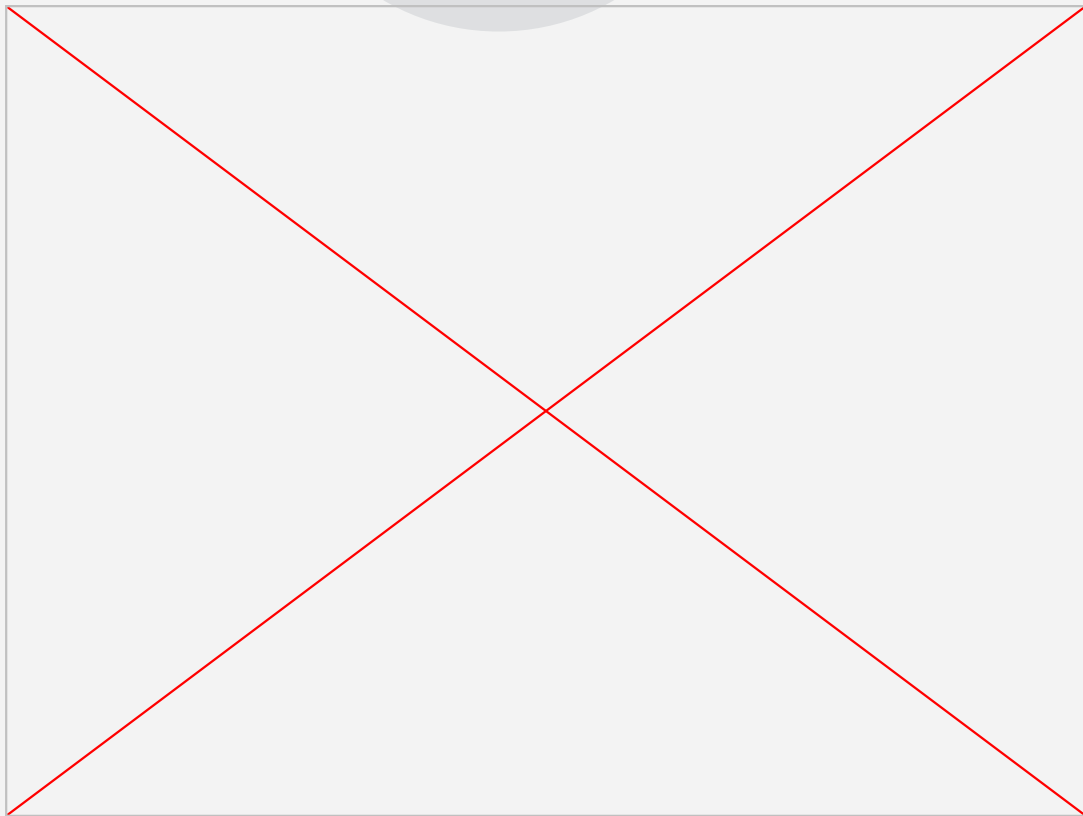
Stephen Curry

Year 2:

2013

Compare Player Shot Locations







Conclusion

- Three-point attempt rate has grown from ~22% in 2004 to over 42% in 2024.
 - And mid-range shots, once 30% of all attempts, now account for less than 12%.
 - Yes! He changed the game.
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